

A Vision-Based Framework for PM2.5 Classification Using Supervised and Self-Supervised Learning

Abstract—Accurate and scalable monitoring of fine particulate matter (PM2.5) is essential due to its severe impacts on human health and environmental sustainability. Conventional PM2.5 monitoring relies on dense sensor networks, which are costly to deploy and provide limited spatial coverage. In this work, we propose a vision-based framework for PM2.5 classification that integrates fully supervised and self-supervised learning paradigms to enable data-efficient air quality assessment from outdoor images. The proposed framework evaluates multiple pretrained deep learning models, including convolutional neural networks and Vision Transformers, under supervised settings and compares them with self-supervised representation learning approaches such as SimCLR, MoCo, BYOL, Masked Autoencoders (MAE), and DINOv2. Experiments conducted on the PM2.5-Vision dataset (11,103 images across multiple air quality categories) demonstrate that self-supervised learning significantly improves label efficiency and robustness, particularly in low-label scenarios. Contrastive methods achieve stronger downstream performance than reconstruction-based approaches, while DINOv2 provides the most discriminative and transferable representations, yielding the highest overall classification accuracy with balanced precision and recall. The proposed framework also analyzes computational efficiency and performance stability across multiple train-test splits. Overall, this study highlights the effectiveness of self-supervised representation learning for scalable and practical vision-based PM2.5 monitoring.

Index Terms—PM2.5 classification, Vision-based air quality monitoring, Self-supervised learning, Deep learning, Representation learning, Image-based pollution analysis.

I. INTRODUCTION

Fine particulate matter (PM2.5) is a major air pollutant with severe impacts on human health, climate, and environmental sustainability. Long-term exposure to elevated PM2.5 levels is linked to respiratory and cardiovascular diseases and increased mortality, making accurate and timely monitoring essential. However, traditional PM2.5 monitoring relies on dense networks of fixed sensor stations that are costly and provide limited spatial coverage.

Early approaches used statistical and machine learning models based on meteorological and sensor data but suffered from poor spatial generalization. With advances in deep learning, vision-based PM2.5 estimation emerged, leveraging outdoor and satellite imagery to infer atmospheric conditions. Initial studies employed pretrained CNNs to extract haze-related features for PM2.5 classification [1], followed by ensemble CNNs [2] and classification schemes aligned with regulatory thresholds [3]. Later works incorporated temporal modeling using recurrent and attention-based networks [4], spatial modeling via graph neural networks [5], and transformer-based architectures that captured global spatial dependencies from

satellite imagery [6]. Self-supervised and semi-supervised methods were also explored to reduce reliance on labeled air quality data [7], [5].

Despite these advances, most vision-based PM2.5 studies rely on fully supervised learning and lack systematic evaluation under low-label conditions. Since obtaining accurate PM2.5 labels is costly, data-efficient learning is critical. Self-supervised learning (SSL) methods such as SimCLR, MoCo, BYOL, and Masked Autoencoders (MAE) have achieved strong performance in computer vision but remain underexplored for PM2.5 estimation.

To address this gap, we propose PM2.5-Vision, a unified framework that evaluates CNNs and Vision Transformers under supervised learning and compares them with representations learned via SimCLR, MoCo, BYOL, MAE, and pretrained DINOv2. The framework examines linear probing, classical classifiers on frozen embeddings, and end-to-end fine-tuning under varying label availability. Results show that contrastive SSL methods perform best in low-label regimes, while DINOv2 provides the strongest overall representations. PM2.5-Vision offers a scalable and generalization-aware solution for vision-based PM2.5 monitoring.

II. LITERATURE REVIEW

Early research on air quality and PM2.5 estimation primarily relied on traditional statistical and machine learning techniques using meteorological and sensor-based measurements. These methods demonstrated reasonable predictive capability but were constrained by sparse monitoring stations and limited spatial coverage, restricting their applicability in large-scale or fine-grained air quality assessment.

With the growth of deep learning, researchers began exploring vision-based PM2.5 estimation using convolutional neural networks (CNNs). One of the earliest studies demonstrated that atmospheric visibility and haze information extracted from outdoor images could be used to classify PM2.5 concentration levels using pretrained CNNs [1]. Although this work established the feasibility of image-driven air quality estimation, its performance was limited by a small dataset collected from a single city and sensitivity to illumination and weather variations. Subsequent work extended this idea by introducing ensemble deep learning frameworks, combining multiple CNN predictors to reduce estimation variance and improve robustness [2]. More recently, multispectral image-based ensemble approaches using drone imagery demonstrated improved PM2.5 detection by exploiting complementary spectral information, highlighting the benefit of ensemble learning for

complex atmospheric conditions [8]. However, these methods often increase computational complexity and are less suitable for lightweight deployment.

Parallel to image-based approaches, researchers investigated threshold-based and classification-oriented formulations of air pollution prediction. Instead of estimating exact PM2.5 values, several studies reformulated the problem as polluted versus non-polluted classification using machine learning models such as logistic regression and gradient-based methods [3]. These early classification approaches aligned well with regulatory standards but often relied on simplified assumptions and limited validation protocols.

As time-series air quality data became more widely available, temporal modeling techniques gained prominence. Recurrent neural networks, including LSTM and BiLSTM architectures, were introduced to capture temporal dependencies in PM2.5 concentration data and improve short-term forecasting performance [4]. Attention mechanisms were later incorporated to emphasize critical time windows and influential variables, leading to more accurate multi-class PM2.5 level prediction and forecasting [4].

To improve spatial generalization and address the sparsity of monitoring stations, recent studies began integrating spatiotemporal modeling and graph-based learning. Graph neural networks (GNNs) were employed to model spatial relationships among air quality monitoring stations, enabling estimation and forecasting at unmonitored locations [5]. Semi-supervised and self-supervised learning strategies were further introduced to reduce reliance on labeled data and enhance robustness under data-scarce conditions [5], [7]. Multimodal self-supervised frameworks that combine remote sensing imagery with auxiliary information further demonstrated the effectiveness of representation learning for air quality estimation without extensive annotations [9].

With advancements in attention-based architectures, researchers explored transformer-based models for air quality estimation. Vision Transformer (ViT)-based frameworks demonstrated strong capability in capturing global spatial dependencies from satellite imagery and remote sensing data. When combined with meteorological variables and temporal modeling components, these approaches achieved improved regression accuracy and stronger spatial generalization, particularly under station-based cross-validation settings [6]. These findings highlight the importance of global attention mechanisms and multimodal data fusion in PM2.5 estimation.

More recently, practical deployment considerations have motivated the development of lightweight and mobile-friendly deep learning models. Image-driven PM2.5 estimation frameworks optimized for mobile devices demonstrated that reasonable prediction accuracy can be achieved with reduced computational cost, enabling scalable and low-cost air quality monitoring solutions [10]. However, these studies also emphasize persistent challenges related to domain shift across devices, environments, and geographic regions.

In parallel, contemporary research has increasingly focused on decision-oriented air quality prediction, where PM2.5 es-

timations are framed as a threshold-exceedance or risk classification task rather than precise numerical regression. Gradient-boosting models and deep classifiers optimized through systematic hyperparameter tuning and explainability techniques such as SHAP have shown strong performance while providing interpretable insights for public health decision-making [11], [4].

Overall, the chronological progression of research indicates a clear evolution from traditional sensor-based models to deep CNN-based vision approaches, followed by ensemble learning, temporal modeling, graph-based spatiotemporal learning, transformer architectures, and self-supervised techniques. These advances collectively demonstrate that robust PM2.5 estimation in real-world settings requires the integration of strong feature extraction, temporal and spatial modeling, multimodal data fusion, and generalization-aware evaluation strategies.

III. METHODOLOGY

We present the overall workflow of the proposed PM2.5-Vision framework, as illustrated in Fig. 1. In this study, we utilize the PM2.5-Vision dataset, which consists of outdoor images associated with corresponding PM2.5 air quality levels. The images were first preprocessed by resizing them to a fixed resolution and normalizing pixel intensities to ensure consistency across samples. The dataset was then divided into training, validation, and testing subsets using stratified sampling to preserve class distribution. Standard data augmentation techniques were applied during training to improve generalization. To systematically analyze image-based PM2.5 classification, we evaluated both fully supervised learning and self-supervised learning (SSL) paradigms. In the supervised setting, multiple deep learning models were trained end-to-end using labeled images with cross-entropy loss, the Adam optimizer, and early stopping to prevent overfitting. In parallel, self-supervised pretext training was conducted on unlabeled PM2.5-Vision images to learn general visual representations without manual annotations. The learned representations were subsequently transferred to downstream classification tasks. For downstream evaluation, we adopted a multi-level strategy including linear probing on frozen features, classical machine learning classifiers trained on extracted embeddings, and end-to-end fine-tuning using labeled data. To assess label efficiency, experiments were performed using varying proportions of labeled samples, enabling evaluation under low-label conditions. Model performance was assessed using accuracy, precision, recall, and F1-score, along with confusion matrix analysis to examine class-wise behavior. Additionally, inference time and computational efficiency were analyzed to evaluate practical deployment feasibility.

A. Dataset

The PM2.5-Vision dataset [12] consists of 11,103 outdoor images associated with different PM2.5 air quality levels, capturing visual characteristics such as atmospheric haze, visibility degradation, and illumination variations under real

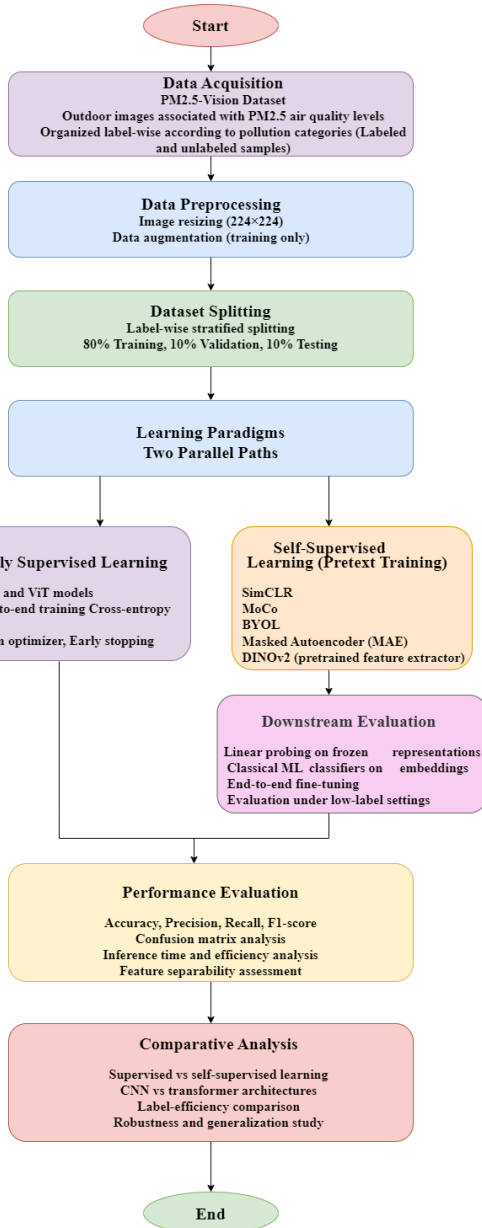


Fig 1. Visualization of the Implementation Pipeline

environmental conditions. Each image corresponds to a specific pollution category derived from air quality measurements, making the dataset suitable for image-based PM2.5 classification. Table I shows the division of PM2.5 air quality classes along with their pollution ranges, visual characteristics, and representative images.

During preprocessing, the dataset was reorganized into a label-wise directory structure according to pollution categories to ensure consistency and ease of access during training. All images were resized to a fixed resolution of 224x224 pixels and normalized to maintain uniform input representation, while data augmentation was applied only during the training phase to improve model generalization. The dataset

TABLE I
PM25 VISION CLASS WITH DESCRIPTION AND IMAGE

Class Type	Description	Image
Good (0-50)	Clear view from inside a vehicle on a quiet paved road. Houses and utility poles line the left side behind a white guardrail, while a grassy embankment runs along the right under an overcast sky.	
Moderate (51-100)	A wide-angle street view of a quiet intersection beside a low commercial building, with empty roads and a partly cloudy sky overhead	
Unhealthy Sensitive (101-150)	A forward-facing dashcam view of a wide intersection with zebra crossings and overhead traffic lights, with trees lining the right side under a cloudy, gray sky.	
Unhealthy (151-200)	A broad, mostly empty multi-lane road stretching through an urban area, with buildings and streetlights on both sides and a few pedestrians and motorcycles in the distance under a hazy sky.	
Very Unhealthy (201-300)	A hazy street scene with several cars and a white van in traffic or parked near commercial buildings, viewed through a dusty or fogged window under bright daylight	
Hazardous (301-10000)	A busy city intersection with heavy mixed traffic (cars, motorcycles, and auto-rickshaws) moving around a road divider. Old commercial buildings with shop signs and large billboards surround the junction under a hazy sky.	

was divided using label-wise stratified sampling into 80% training, 10% validation, and 10% testing subsets, preserving class distribution across splits. This processed dataset supports both supervised learning and self-supervised pretext training, where labels are used only during supervised and downstream evaluation stages.

B. Model Description

1) **Momentum Contrast (MoCo)**: Momentum Contrast (MoCo) is a self-supervised learning framework used in this work to learn discriminative visual representations from the PM2.5-Vision dataset without relying on PM2.5 labels during training. All input images were resized to 224x224 and passed through a convolutional backbone network that acts as a feature encoder. For each image, two different augmented versions were generated, allowing the model to observe the same atmospheric scene under varied visual conditions. These

two views were processed by a query encoder and a key encoder, respectively, both having identical architectures.

To ensure stable representation learning, the key encoder was updated using a momentum-based moving average of the query encoder parameters instead of direct backpropagation. This design enabled the model to maintain consistent feature representations over training iterations. MoCo also employed a fixed-size memory queue that stored feature embeddings from previously processed images, which allowed the model to compare the current image representation against a large number of negative samples without increasing batch size. Through contrastive optimization, the model learned to distinguish similar atmospheric patterns from dissimilar ones, effectively capturing haze density, visibility degradation, and illumination variations present in PM2.5-Vision images.

After self-supervised training, the pretrained MoCo encoder was reused for PM2.5 classification by extracting feature representations from labeled images. These features were evaluated using linear classifiers, traditional machine learning models, and end-to-end fine-tuning to assess their effectiveness under different label availability settings. The MoCo-based representations demonstrated strong transferability and improved performance, particularly in scenarios with limited labeled data, highlighting the suitability of contrastive self-supervised learning for image-based PM2.5 analysis.

2) **Bootstrap Your Own Latent (BYOL):** Bootstrap Your Own Latent (BYOL) is a self-supervised learning approach adopted in this work to learn robust visual representations from the PM2.5-Vision dataset without using PM2.5 labels during training. All input images were resized to $224 \times 224 \times 3$ and processed using a convolutional backbone network that acts as a feature encoder. For each PM2.5-Vision image, two differently augmented views were generated to represent the same atmospheric scene under varying visual conditions such as illumination and haze intensity. These two views were passed through an online network and a target network, both sharing the same encoder architecture.

The online network was trained to predict the latent representation produced by the target network, while the target network parameters were updated using an exponential moving average of the online network parameters. This momentum-based update mechanism stabilized training and prevented representation collapse, even without the use of negative samples. Through this teacher–student learning process, BYOL encouraged the model to learn invariant visual features related to haze density, visibility reduction, and contrast variations present in PM2.5-Vision images.

After self-supervised training, the pretrained BYOL encoder was transferred to the PM2.5 classification task by extracting feature representations from labeled images. These features were evaluated using linear probing, classical machine learning classifiers, and end-to-end fine-tuning to assess representation quality and classification performance under limited labeled data. The BYOL-based representations demonstrated strong stability and generalization, making the approach effective for learning atmospheric visual patterns in outdoor PM2.5 scenes.

3) **Simple Framework for Contrastive Learning of Visual Representations (SimCLR):** Simple Framework for Contrastive Learning of Visual Representations (SimCLR) is employed in the PM2.5-Vision framework as a contrastive self-supervised learning approach to learn discriminative visual features from outdoor images without using PM2.5 labels during training. All input images were resized to $224 \times 224 \times 3$ and augmented twice to generate two correlated views of the same scene, allowing the model to observe identical atmospheric conditions under different visual transformations. These augmented image pairs were processed using a convolutional backbone encoder followed by a projection head that maps features into a latent representation space.

During training, SimCLR maximizes the similarity between representations of the two augmented views originating from the same image while minimizing similarity with representations from other images within the batch using a contrastive loss function. This encourages the model to learn invariant visual features related to haze density, visibility reduction, and illumination changes present in PM2.5-Vision images. By relying on strong data augmentations and contrastive objectives, SimCLR effectively captures meaningful atmospheric patterns without requiring explicit supervision.

After self-supervised training, the pretrained SimCLR encoder was transferred to the PM2.5 classification task by extracting feature representations from labeled images. These representations were evaluated through linear probing, classical machine learning classifiers, and end-to-end fine-tuning to assess representation quality and classification performance under varying levels of labeled data availability. The SimCLR-based representations demonstrated strong transferability and effectiveness, particularly in low-label scenarios, highlighting the suitability of contrastive self-supervised learning for image-based PM2.5 analysis.

4) **DINOv2 (Self-Distillation with No Labels v2):** DINOv2 is employed in the PM2.5-Vision framework as a large-scale self-supervised Vision Transformer used for extracting high-quality visual representations without requiring task-specific pretext training. All PM2.5-Vision images were resized to $224 \times 224 \times 3$ and divided into fixed-size patches, which were then processed by the transformer encoder using self-attention mechanisms. Unlike other self-supervised approaches trained directly on the PM2.5-Vision dataset, DINOv2 leverages pre-trained weights learned from large and diverse image collections, enabling strong generalization capability.

The self-attention mechanism in DINOv2 allows the model to capture long-range dependencies and global spatial relationships across the entire image, which is particularly effective for modeling atmospheric patterns such as haze distribution and visibility variation that span large regions of outdoor scenes. When applied to the PM2.5-Vision dataset, DINOv2 produces highly discriminative feature embeddings that preserve semantic structure and visual consistency across pollution categories.

After feature extraction, the DINOv2 representations were transferred to the PM2.5 classification task by training lightweight classifiers on top of the frozen embeddings and

evaluating their performance using labeled data. Additional analysis, including feature separability visualization, was performed to assess the quality of the learned representations. The strong performance of DINOv2 highlights the effectiveness of large-scale self-supervised pretraining for image-based PM2.5 analysis, particularly in scenarios with limited labeled data.

5) **Masked Autoencoder (MAE)**: Masked Autoencoder (MAE) is employed in the PM2.5-Vision framework as a reconstruction-based self-supervised learning approach to learn meaningful visual representations from outdoor images without using PM2.5 labels during training. All input images were resized to 224×224×3 and divided into fixed-size patches, where a large portion of the patches was randomly masked before being fed into the encoder. Only the visible patches were processed by the encoder network, forcing the model to learn compact and informative representations from incomplete visual information.

During training, the encoder learned to capture global contextual cues related to haze distribution, visibility reduction, and atmospheric texture by reconstructing the masked image regions through a lightweight decoder. This masking strategy encouraged the model to focus on high-level scene understanding rather than local pixel-level details, which is particularly suitable for PM2.5-Vision images where pollution effects are spread across large regions of the scene. The MAE model was trained for a fixed number of epochs, and the pretrained encoder weights were saved for transfer learning.

After self-supervised training, the MAE encoder was reused for PM2.5 classification by extracting feature representations from labeled images. These representations were evaluated using linear probing, classical machine learning classifiers, and end-to-end fine-tuning to assess their effectiveness under limited labeled data. The MAE-based representations demonstrated strong global feature learning capability, making the approach effective for capturing large-scale atmospheric patterns present in PM2.5-Vision images.

6) **Pretrained Supervised Deep Learning Models**: As an initial stage of the PM2.5-Vision framework, several pretrained deep learning models were employed as fully supervised baselines to evaluate image-based PM2.5 classification performance. All input images were resized to 224×224×3 and normalized to ensure consistency across models. These models were initialized with ImageNet-pretrained weights, allowing them to leverage rich visual representations learned from large-scale image datasets, and were subsequently fine-tuned using labeled PM2.5-Vision images.

The convolutional neural network (CNN)-based models, including ResNet-50, VGG-16, MobileNetV3, and GoogLeNet, learn hierarchical spatial features through stacked convolutional layers. Early layers capture low-level patterns such as edges and color variations, while deeper layers learn higher-level semantic features related to haze density, contrast reduction, and scene clarity. Residual connections in ResNet-50 facilitate deeper feature learning by mitigating vanishing gradient issues, while GoogLeNet employs inception modules to extract multi-scale features efficiently. MobileNetV3 uti-

lizes lightweight depthwise separable convolutions, making it computationally efficient and suitable for resource-constrained environments, whereas VGG-16 provides a deeper but straightforward convolutional structure for robust feature extraction.

In addition to CNN-based models, the Vision Transformer (ViT-B/16) was used to explore transformer-based learning for PM2.5 classification. ViT divides each image into fixed-size patches and processes them as a sequence of tokens using self-attention mechanisms. This enables the model to capture global contextual relationships across the entire image, which is beneficial for modeling large-scale atmospheric patterns such as widespread haze and visibility degradation. For all pretrained supervised models, the final classification layers were modified to match the number of PM2.5 pollution categories. The models were trained end-to-end using cross-entropy loss, the Adam optimizer, and early stopping based on validation performance to prevent overfitting. To analyze robustness under varying data availability, each pretrained model was evaluated across nine different train-validation-test split ratios, enabling a comprehensive assessment of performance stability. These supervised models establish strong baseline performance and serve as a reference point for comparison with the self-supervised representation learning approaches applied later in the PM2.5-Vision framework.

C. Evaluation Metrics

To comprehensively evaluate the performance of the proposed PM2.5-Vision framework, multiple standard classification metrics were employed, including overall accuracy, precision, recall, and F1-score. Overall accuracy measures the proportion of correctly classified samples across all PM2.5 categories and is defined as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad (1)$$

where TP , TN , FP and FN denote true positives, true negatives, false positives, and false negatives, respectively. Accuracy provides a general indication of model performance across all classes.

Precision reflects the model's ability to correctly identify samples belonging to a specific PM2.5 class and is given by

$$\text{Precision} = \frac{TP}{TP + FP}. \quad (2)$$

Recall measures how effectively the model captures all relevant samples of a given PM2.5 class and is defined as

$$\text{Recall} = \frac{TP}{TP + FN}. \quad (3)$$

The F1-score, defined as the harmonic mean of precision and recall, provides a balanced assessment of classification performance and is particularly useful in the presence of class imbalance:

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

Together, these evaluation metrics ensure a fair and robust assessment of both supervised and self-supervised models across different train–test split ratios.

IV. RESULTS AND DISCUSSION

A. Pretrained supervised models across 9 train:test splits

Across all pretrained models, performance follows a clear and expected trend, as summarized in Table II: accuracy and F1 scores are highest when more training data are available (90:10, 80:20, 70:30) and steadily decline as the training portion becomes small (e.g., 20:80 and 10:90). This confirms that PM2.5 classification from outdoor images is a data-dependent task, where models benefit strongly from seeing diverse haze/visibility conditions during training.

Among the supervised baselines, EfficientNet-B0 consistently provides the strongest and most stable results across nearly all splits, peaking at 68.26% (80:20) and maintaining comparatively better performance even when the train set shrinks. This suggests EfficientNet’s parameter-efficient scaling and feature extraction are well-suited for PM2.5 visual cues. ResNet-50 and VGG-16 form a second tier with similar behavior (best around 66–67% at high-train splits), but VGG-16 carries the highest computational cost (30.93 GFLOPs), making it less attractive if deployment efficiency is important. In contrast, MobileNetV3 achieves competitive accuracy at higher-train splits (best 65.53%) while requiring very low computation (0.47 GFLOPs), indicating the best trade-off for lightweight deployment.

ViT-B/16 performs reasonably when training data are sufficient (best 63.30% at 80:20), but its accuracy degrades more sharply under extreme low-training splits (e.g., 39.45% at 10:90). This pattern is consistent with transformer-based vision models, which often require more data to stabilize learning compared to CNNs. Finally, GoogLeNet shows significantly weaker results across all splits (best 37.99%) despite moderate compute (3.02 GFLOPs), indicating that in your PM2.5-Vision setting it fails to learn sufficiently discriminative haze/visibility representations, or its optimization may be less stable compared to the other backbones.

Overall, the supervised experiments show that the dataset contains learnable visual patterns for PM2.5 levels, and the most reliable supervised baseline is EfficientNet-B0, while the most deployment-friendly option is MobileNetV3 due to its strong accuracy-to-compute balance.

B. Self-Supervised Learning Results (MoCo, BYOL, SimCLR, MAE, DINOv2)

The self-supervised learning results, summarized in Table III, reveal a clear hierarchy in representation quality when evaluated using overall accuracy, precision, recall, and F1-score. Among all SSL approaches, DINOv2 consistently achieves the strongest performance, with the MLP classifier reaching the highest overall accuracy (65.90%), along with balanced precision (0.6920), recall (0.6670), and F1-score (0.6780) under the 80:20 split. This balanced improvement across all metrics indicates that large-scale self-supervised

TABLE II
PRE-TRAINED MODEL PERFORMANCE

Model	Train:Test	Accuracy (%)	Precision	Recall	F1-score	GFLOPs
ViT (Vision Transformer)	90:10	60.76	0.6143	0.6511	0.6260	22.57
	80:20	63.30	0.6868	0.6462	0.6611	22.57
	70:30	61.30	0.6517	0.6386	0.6428	22.57
	60:40	59.70	0.6119	0.6353	0.6178	22.57
	50:50	60.84	0.6375	0.6416	0.6227	22.57
	40:60	51.58	0.5985	0.5114	0.5393	22.57
	30:70	50.28	0.5066	0.5249	0.5112	22.57
	20:80	50.08	0.5401	0.5151	0.5221	22.57
	10:90	39.45	0.4714	0.3661	0.3848	22.57
EfficientNet-B0	90:10	67.60	0.7158	0.7252	0.7192	0.83
	80:20	68.26	0.7068	0.7270	0.7140	0.83
	70:30	66.17	0.6932	0.6920	0.6913	0.83
	60:40	63.03	0.6600	0.6621	0.6599	0.83
	50:50	62.84	0.6565	0.6560	0.6550	0.83
	40:60	62.08	0.6598	0.6527	0.6556	0.83
	30:70	58.21	0.6134	0.6176	0.6134	0.83
	20:80	53.43	0.5855	0.5492	0.5603	0.83
	10:90	47.45	0.4728	0.5000	0.4841	0.83
ResNet-50	90:10	67.60	0.6992	0.6934	0.6960	8.26
	80:20	65.30	0.6736	0.6805	0.6719	8.26
	70:30	64.50	0.6791	0.6613	0.6669	8.26
	60:40	61.70	0.6600	0.6433	0.6480	8.26
	50:50	61.67	0.6480	0.6446	0.6455	8.26
	40:60	60.93	0.6470	0.6311	0.6329	8.26
	30:70	59.31	0.6299	0.6128	0.6200	8.26
	20:80	54.75	0.6138	0.5522	0.5740	8.26
	10:90	48.25	0.5081	0.4852	0.4888	8.26
VGG16	90:10	66.10	0.6880	0.6935	0.6890	30.93
	80:20	65.20	0.7004	0.6712	0.6814	30.93
	70:30	64.70	0.6736	0.6735	0.6732	30.93
	60:40	62.60	0.6656	0.6560	0.6579	30.93
	50:50	60.00	0.6242	0.6198	0.6205	30.93
	40:60	57.00	0.5962	0.5947	0.5830	30.93
	30:70	56.34	0.6284	0.5454	0.5704	30.93
	20:80	53.21	0.5803	0.5268	0.5417	30.93
	10:90	46.60	0.4802	0.4453	0.4494	30.93
GoogLeNet	90:10	28.35	0.5556	0.2612	0.2314	3.02
	80:20	36.06	0.5071	0.3292	0.3233	3.02
	70:30	28.13	0.4834	0.2558	0.2189	3.02
	60:40	35.77	0.4791	0.3293	0.3305	3.02
	50:50	34.85	0.5416	0.3312	0.3406	3.02
	40:60	37.99	0.5600	0.3278	0.3339	3.02
	30:70	28.97	0.5299	0.2506	0.2210	3.02
	20:80	30.10	0.5488	0.2724	0.2642	3.02
	10:90	27.70	0.5479	0.2350	0.2160	3.02

pretraining produces highly transferable and well-separated feature representations, enabling effective PM2.5 category discrimination using a lightweight classifier.

Among the SSL methods trained directly on the PM2.5-Vision dataset, BYOL demonstrates the most stable and competitive performance. With the SVM classifier, BYOL achieves 62.55% accuracy, while also maintaining strong precision (0.6908), recall (0.6403), and F1-score (0.6605) under the 70:30 split. These results suggest that BYOL learns structured and consistent feature embeddings with clear class boundaries, making them well-suited for margin-based classifiers. SimCLR follows closely, where the SVM classifier attains 60.20% accuracy, with comparatively high precision (0.6672) and balanced recall and F1-score, indicating effective but slightly less separable representations than BYOL. MoCo shows its best performance with the Random Forest classifier, achieving 58.87% accuracy, alongside improved precision (0.6778), recall (0.5974), and F1-score (0.6249), suggesting that MoCo embeddings benefit from non-linear decision boundaries rather than purely linear separation.

In contrast, MAE yields the weakest performance across all evaluation metrics, with its best result obtained using the MLP classifier at 48.06% accuracy, accompanied by

TABLE III
DOWNSTREAM CLASSIFICATION PERFORMANCE OF DIFFERENT SSL
METHODS USING VARIOUS CLASSIFIERS .

Model	Train:Test	Classifier	Accuracy (%)	Precision	Recall	F1-score
MoCo	80:20	MLP	0.5437	0.5733	0.5825	0.5767
		SVM	0.4959	0.6213	0.4656	0.4988
		DT	0.4149	0.4421	0.4383	0.4400
		RF	0.5887	0.6778	0.5974	0.6249
BYOL	80:20	MLP	0.5918	0.6365	0.6277	0.6161
		SVM	0.6255	0.6908	0.6403	0.6605
		DT	0.4418	0.4517	0.4518	0.4516
		RF	0.6122	0.6998	0.6152	0.6450
SimCLR	80:20	MLP	0.5906	0.6672	0.6234	0.6252
		SVM	0.6020	0.6672	0.6148	0.6347
		DT	0.4328	0.4614	0.4546	0.4576
		RF	0.5906	0.6721	0.6023	0.6261
DINOv2	80:20	MLP	0.6590	0.6920	0.6670	0.6780
		SVM	0.5810	0.6130	0.5920	0.5990
		DT	0.3610	0.3410	0.3460	0.3430
		RF	0.5250	0.6680	0.4750	0.5090
MAE	80:20	MLP	0.4806	0.5282	0.4969	0.5054
		SVM	0.3114	0.2074	0.2394	0.2021
		DT	0.3186	0.3125	0.3126	0.3119
		RF	0.4752	0.5823	0.4475	0.4768

relatively lower precision, recall, and F1-score. This trend indicates that reconstruction-based self-supervised learning is less aligned with the PM2.5 classification objective in this setting. Although MAE effectively captures global scene structure, it does not sufficiently emphasize class-discriminative atmospheric features such as haze intensity and visibility degradation, resulting in reduced downstream separability, particularly when shallow classifiers are used.

Overall, the SSL results demonstrate that contrastive and teacher–student-based approaches outperform reconstruction-based learning, and that large-scale pretrained self-supervised models (DINOv2) provide the most robust and balanced improvements across all evaluation metrics. These findings highlight the importance of representation quality in image-based PM2.5 classification, especially under limited labeled data scenarios.

V. CONCLUSION

This paper presented a vision-based framework for PM2.5 classification that integrates fully supervised and self-supervised learning strategies to enable scalable and data-efficient air quality monitoring. Using the PM2.5-Vision dataset, extensive experiments were conducted across multiple pretrained deep learning architectures and self-supervised learning methods to analyze classification accuracy, robustness under varying train–test splits, and label efficiency.

The experimental results demonstrate that pretrained supervised models can effectively capture visual cues related to haze and visibility, with modern CNN architectures providing strong and stable baseline performance. However, their performance degrades as labeled data become scarce. In contrast, self-supervised learning approaches significantly reduce dependence on labeled data by learning transferable visual representations from unlabeled images. Among the evaluated SSL methods, contrastive and teacher–student-based approaches

show superior robustness compared to reconstruction-based learning, while the large-scale pretrained DINOv2 model provides the most discriminative and generalizable representations for PM2.5 classification.

Overall, the findings highlight the importance of representation learning for vision-based air quality assessment and demonstrate that self-supervised learning is a promising solution for real-world PM2.5 monitoring scenarios where labeled data are limited. Future work will focus on extending the framework to regression-based PM2.5 estimation, incorporating multimodal data such as meteorological variables, and deploying lightweight models for real-time and edge-based air quality monitoring.

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